Depression Analysis

Lifetime Diagnosis of Depression

Research Question:

How does the association between reported lifetime depression prevalence and reported subjective cognitive decline differ between adults aged 50–64 and adults aged 65 years or older?

Dataset:

This analysis uses the CDC Alzheimer's Disease and Healthy Aging dataset. The two primary topics analyzed were **Lifetime diagnosis of depression** and **Subjective cognitive decline or memory loss among older adults**. The analysis focuses on adults aged **50–64 years** and **65 years or older**. Depression and SCD estimates were matched by year, location, and age group.

Initial Depression Comparison:

Does reported lifetime depression prevalence differ between adults aged 50–64 and adults aged 65 years or older?

Null Hypothesis

H_0 - There is no statistically significant difference in reported lifetime depression prevalence between the two age groups.

Alternative Hypothesis

H_1 - There is a statistically significant difference.

Table 1: Descriptive Statistics

Age group	N	Mean	Median	SD	Q1	Q3	IQR	Outliers

50–64	463	20.730%	20.500%	4.046	18.050	23.250	5.200	6(1.30%)
65+	460	14.779%	14.650%	2.562	13.100	16.500	3.400	2(0.43%)

The descriptive statistics compare reported lifetime depression prevalence between adults aged 50–64 and adults aged 65 years or older. The 50–64 group had a higher mean reported depression prevalence (20.730%) than the 65+ group (14.779%). The median was also higher for the 50–64 group (20.500%) compared with the 65+ group (14.650%). The 50–64 group had a standard deviation of 4.046, while the 65+ group had a standard deviation of 2.562, indicating greater variation in reported depression prevalence among the 50–64 estimates.

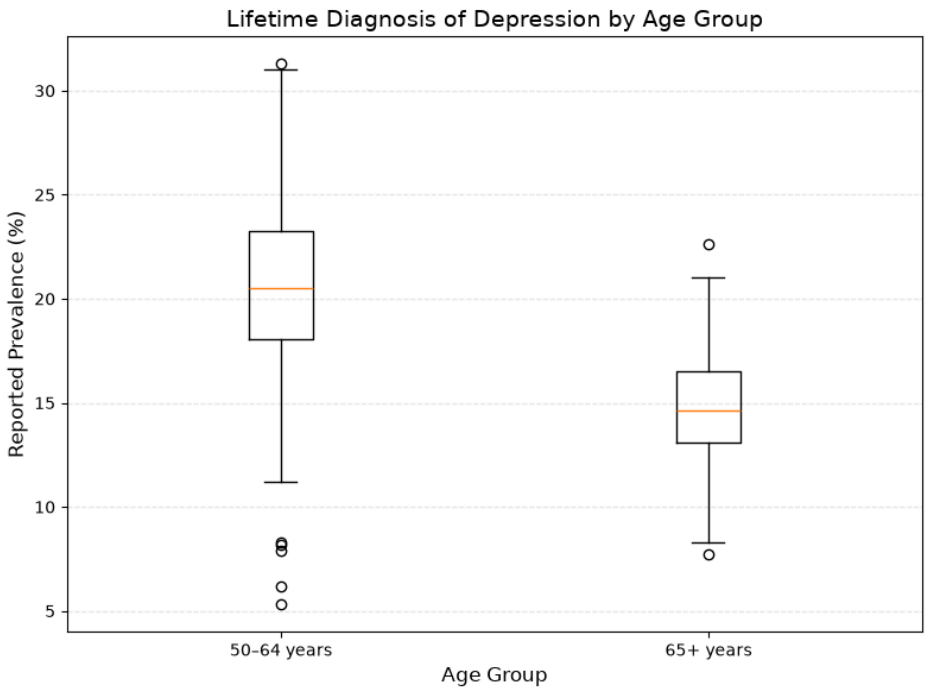


Figure 1. The boxplot compares the distribution of reported lifetime depression prevalence between adults aged 50–64 and adults aged 65 years or older. The 50–64 group has a higher median prevalence (20.50%) than the 65+ group (14.65%) and shows greater variability. Several statistical outliers are present in both groups, with more outliers in the 50–64 group.

Welch's t-Test:

A Welch's two-sample t-test was used to compare reported depression prevalence between the two age groups.

- t = **26.7160**
- p < **0.000001**

The difference was statistically significant, so the null hypothesis was rejected. Reported depression prevalence was higher among adults aged 50–64 than adults aged 65 years or older.

Outlier Analysis

Age group	Q1	Q3	IQR	Lower bound	Upper bound	Outliers
50–64	18.050	23.250	5.200	10.250	31.050	6
65+	13.100	16.500	3.400	8.000	21.600	2

IQR-based outlier detection identified 6 observations in the 50–64 group and 2 observations in the 65+ group. These observations were treated as statistical outliers rather than assumed to be erroneous data.

Outlier Sensitivity

Age group	N	Mean
50–64	457	20.856%
65+	458	14.778%

Welch test:

- t = **28.6076**
- p < **0.000001**

The significant difference remained after removing IQR outliers, suggesting that the age-group difference was not dependent on the identified extreme observations.

Depression and SCD Comparison

Key Question:
Is reported depression prevalence associated with reported SCD, and does this association differ between the two age groups?

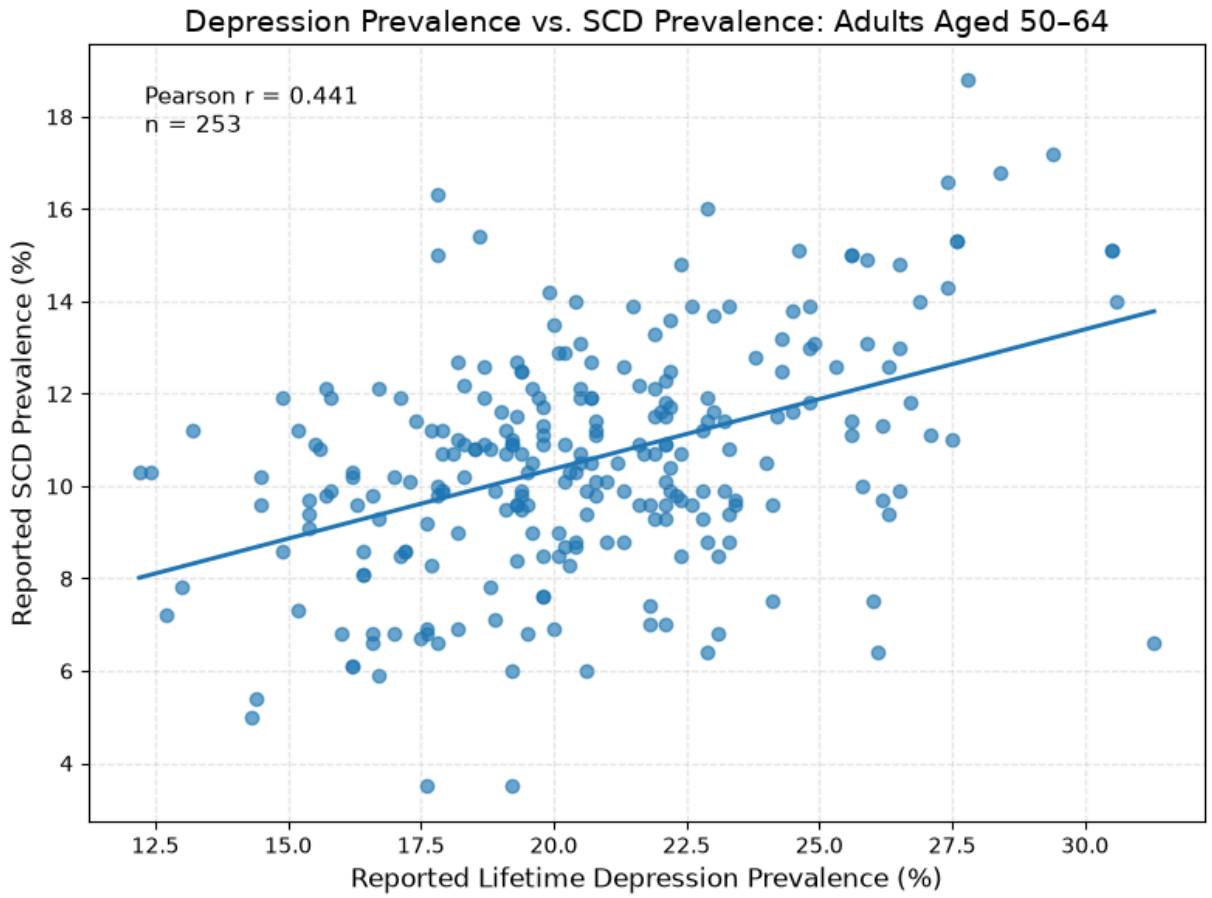


Figure 2. Depression Prevalence vs. SCD Prevalence Among Adults Aged 50–64.

The scatterplot shows a positive association between reported lifetime depression prevalence and reported subjective cognitive decline (SCD) prevalence among adults aged 50–64. Across 253 matched year-location observations, the Pearson correlation was **r = 0.441**, indicating a moderate positive association. The upward-sloping regression line is consistent with this relationship.

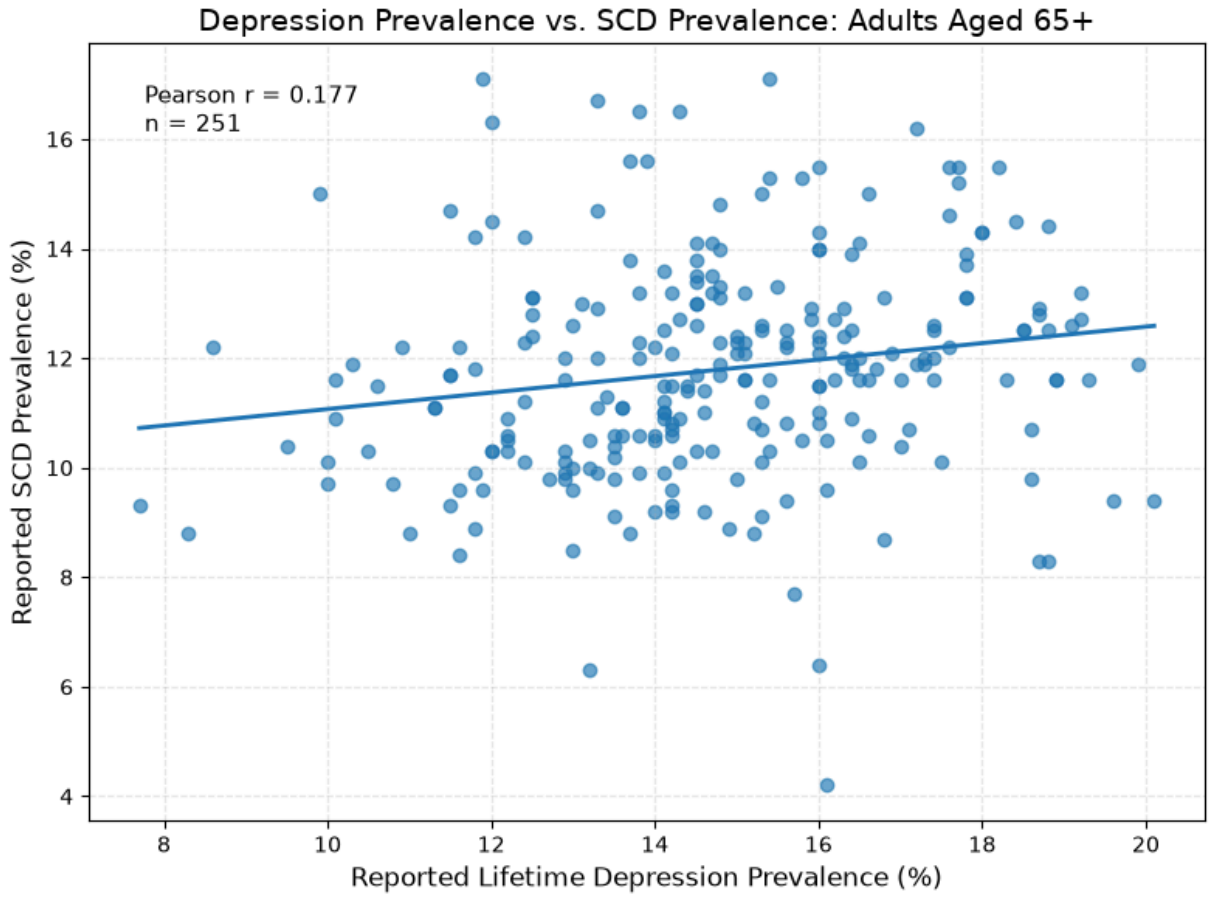


Figure 3. Depression Prevalence vs. SCD Prevalence Among Adults Aged 65+. The scatterplot shows a positive but weaker association between reported lifetime depression prevalence and reported SCD prevalence among adults aged 65 years or older. Across 251 matched year-location observations, the Pearson correlation was r = 0.177. The association was statistically significant (p = 0.00492).

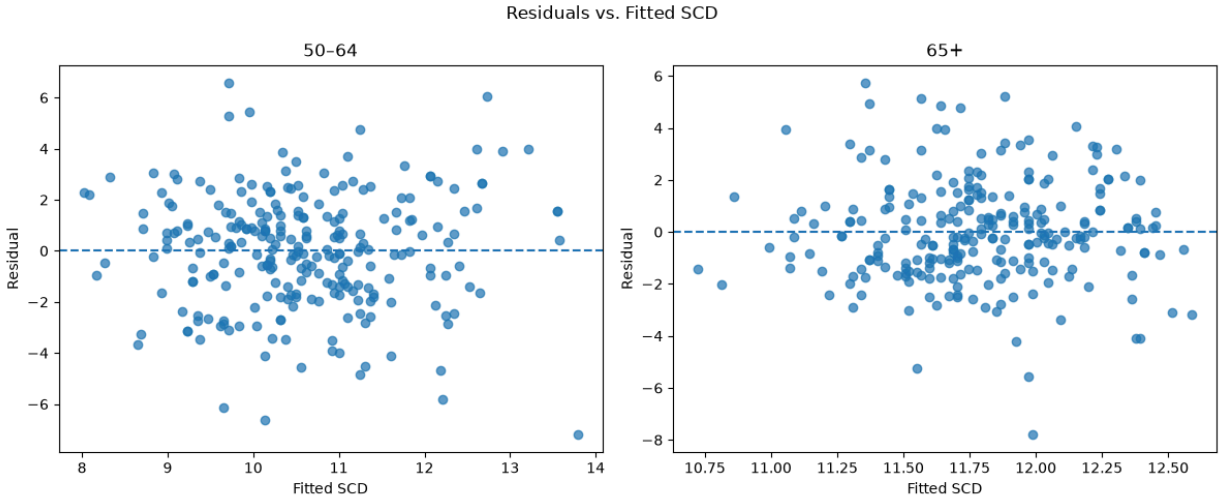


Figure 4. Residuals vs. Fitted SCD

The residual plots provide evidence about whether a linear model is an appropriate representation of the relationship between depression prevalence and SCD prevalence.

50–64:

The residuals are generally distributed around the zero line across the range of fitted SCD values. There is no obvious curved pattern, which suggests that the relationship between depression prevalence and SCD prevalence is reasonably consistent with a linear form.
There are several relatively large positive and negative residuals, particularly at the extremes. However, these observations do not appear to form a systematic pattern. The spread of the residuals also does not show an obvious funnel-shaped pattern, providing no strong visual indication of substantial heteroscedasticity.

65+:

The 65+ residuals are also generally centered around zero, with no clear systematic curvature. This again supports using a linear relationship to describe the association between depression prevalence and SCD prevalence.
There are several larger residuals, including some fairly extreme negative and positive values. As with the 50–64 group, however, they do not appear to create an obvious systematic pattern.

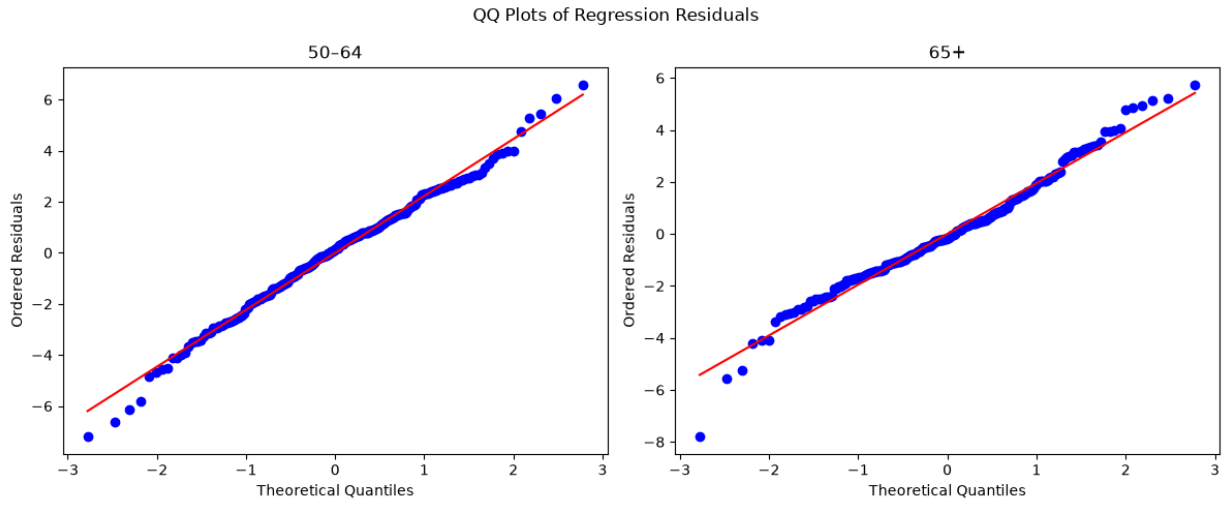


Figure 4. QQ plots

The QQ plots assess whether the regression residuals approximately follow a normal distribution.

50–64:

The residuals follow the reference line quite closely through most of the distribution. The greatest deviations occur in the lower and upper tails.
This suggests that the residuals are approximately normally distributed in the central portion, with some departures from normality among the most extreme observations.

65+:

The same general pattern appears for the 65+ group. Most of the residuals follow the reference line reasonably closely, particularly around the center of the distribution.
There are more noticeable deviations toward the tails, including several extreme residuals.
Therefore, the residual distribution is not perfectly normal, but the main body of the distribution is reasonably consistent with normality.

Conclusion:

Regression diagnostics were conducted to evaluate the suitability of linear regression for the relationship between depression prevalence and SCD prevalence. Residuals were generally distributed around zero without an obvious systematic nonlinear pattern in either age group, supporting the use of a linear model. The QQ plots showed that residuals followed the theoretical normal distribution reasonably closely through the central portion of both distributions, although deviations were present in the tails. These results provide visual support for the linear regression models, while indicating that extreme residuals and departures from normality should be considered when interpreting the results.